

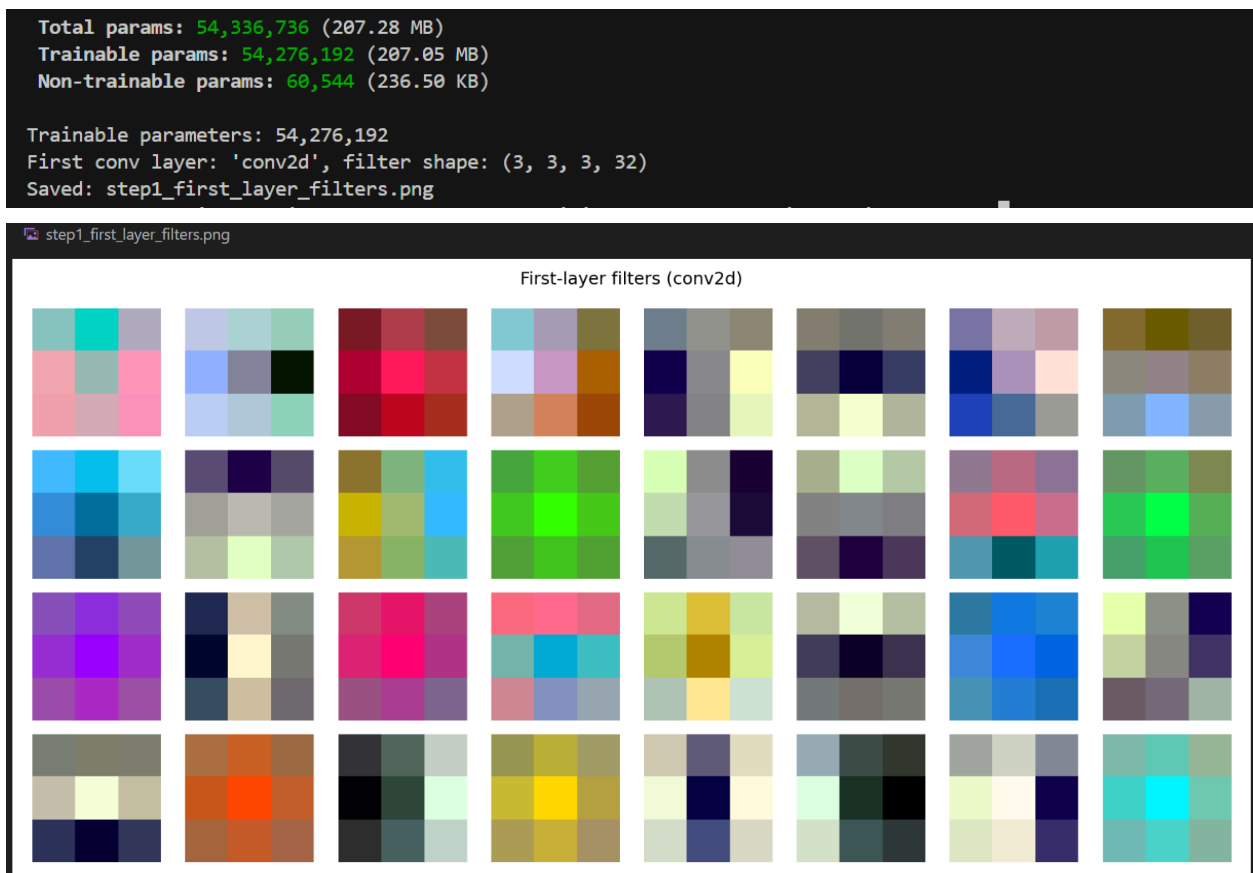
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CS410 computer vision and deep learning assignment 3

The assignment applies transfer learning to a pretrained InceptionResNetV2 CNN to train cats vs dogs. This pretrained convolutional base is frozen and a new classification head is trained on top of it. And I used the dataset of 4000 training images and 1000 test images for each class, dogs and cats.

To load the pretrained model, I used the `InceptionResNetV2(weights = "imagenet", include_top=False, input_shape=(150,150,3))`. This gave me the total params, trainable params, and non-trainable params that are shown in the image below.



The first convolutional layer ('conv2d') has a filter shape of (3, 3, 3, 32); that's 32 filters with a 3 by 3 kernel, with 3 color channels. This gives me the simplest color-gradient patches rather than some object. This is expected of the first layer, doesn't tell us much.

Loading the preprocessing dataset, which I got from my downloaded folder. I needed these steps:

Resize all images to 150 by 150 by 3

Pixel scaling, InceptionResNetV2 built in preprocess_input function

Resulting pixel range, approximately [-1,1]

The preprocess_input function uses the same pixel scaling as the InceptionResNetV2 pretrained on the ImageNet weights that it originally trained with.

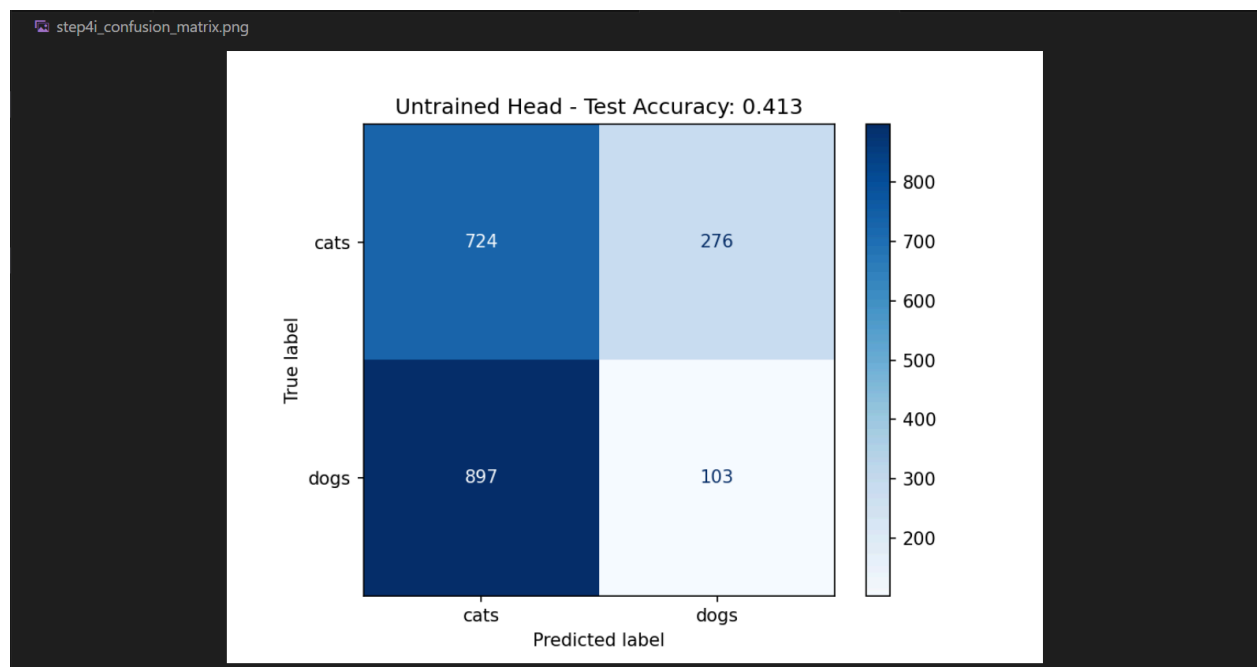
For the transfer head, I needed a new sequential model that I got from stacking a flatten layer, a dense layer, and a final dense output layer on top of the base. This gave me these:

```
Total params: 57,876,193 (220.78 MB)
Trainable params: 57,815,649 (220.55 MB)
Non-trainable params: 60,544 (236.50 KB)

After freezing -- trainable: 3,539,457 frozen: 54,336,736
```

The freezing reduced the trainable parameters by over 90%. And only the newly added classification head will be updated during training.

Before training the model, I evaluated it with this confusion matrix. The pretrained creatures do not default to a neutral 50/50 split.



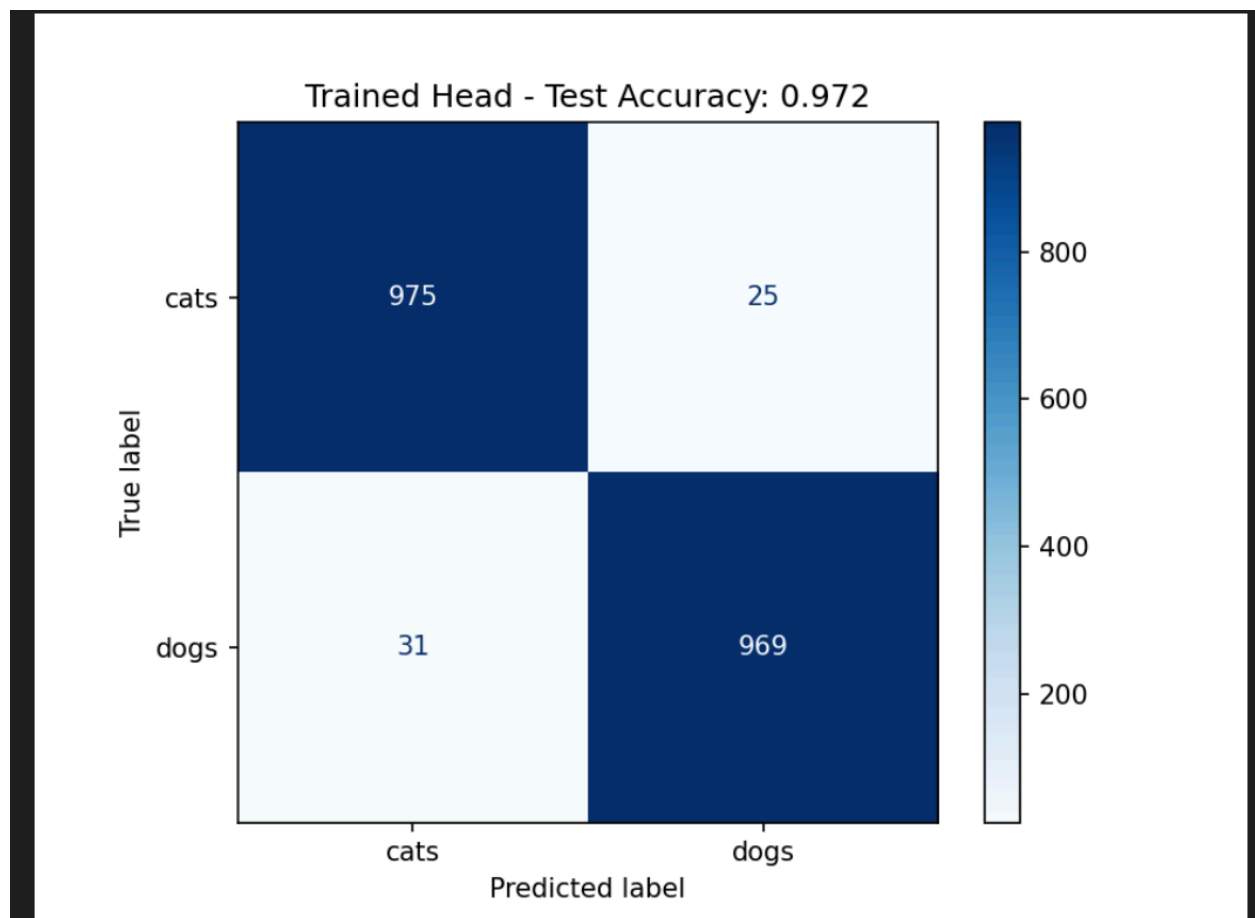
This is what I got when the transfer model was compiled with the Adam optimizer and binary cross-entropy loss. It was trained over 10 epochs.

Per-epoch test loss:

```
epoch 1: train_loss=0.5883 test_loss=0.3386
epoch 2: train_loss=0.1606 test_loss=0.1242
epoch 3: train_loss=0.0430 test_loss=0.1068
epoch 4: train_loss=0.0318 test_loss=0.1231
epoch 5: train_loss=0.0180 test_loss=0.1008
epoch 6: train_loss=0.0130 test_loss=0.1278
epoch 7: train_loss=0.0187 test_loss=0.1112
epoch 8: train_loss=0.0246 test_loss=0.1270
epoch 9: train_loss=0.0163 test_loss=0.1350
epoch 10: train_loss=0.0055 test_loss=0.1212
```

Test accuracy (trained head): 0.9720

Saved: step4ii_confusion_matrix.png



The test accuracy rose from 41.35% to 97.20% after 10 epochs. Confirming the pretrained ImageNet features are highly informative for distinguishing cats from dogs.

The same transfer learning exercise was done again with the sub-network of the pretrained InceptionRexNetV2, constructed by keeping only the first 151 layers. This reduces the frozen base, using more generic features.

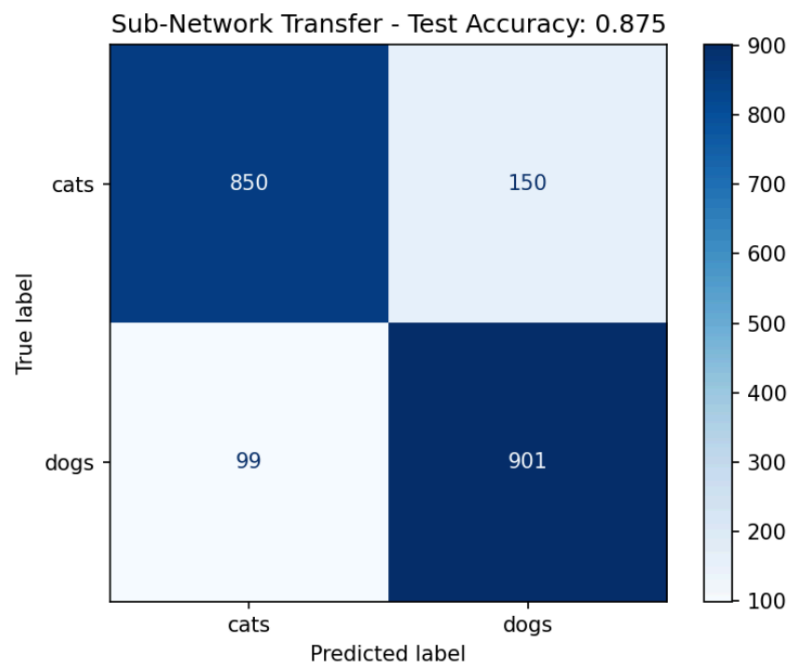
Per-epoch test loss (sub-network):

```
epoch 1: train_loss=1.2124 test_loss=0.4044
epoch 2: train_loss=0.4903 test_loss=0.3832
epoch 3: train_loss=0.3431 test_loss=0.3966
epoch 4: train_loss=0.2110 test_loss=0.8067
epoch 5: train_loss=0.1752 test_loss=0.3987
epoch 6: train_loss=0.0875 test_loss=0.5439
epoch 7: train_loss=0.0887 test_loss=0.6308
epoch 8: train_loss=0.2374 test_loss=0.4419
epoch 9: train_loss=0.0645 test_loss=0.4167
epoch 10: train_loss=0.0294 test_loss=0.4021
```

Test accuracy (sub-network, trained head): 0.8755

Saved: step4iii_confusion_matrix.png

step4iii_confusion_matrix.png



This sub-network reached a final test accuracy of 87.55%, almost 10% lower than the full network. Having about 6 times more trainable parameters to try to compensate for the smaller feature extractor. The per-epoch was more noisy. The training was noticeably faster than the full network.